

Beyond Detection Parity: Localization Parity in Pedestrian Detection

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Abstract. Pedestrian detection is a safety-critical component of autonomous driving and intelligent surveillance, where equitable performance across demographic groups is essential for trustworthy deployment. Fairness research in this domain has evaluated performance almost exclusively through detection recall, that is, whether a person is found at all, leaving the precision of the predicted bounding box unexamined. Because existing audits reduce localization to a binary hit-or-miss threshold, systematic disparities in how tightly detectors localize different groups remain invisible, even though loose boxes directly compromise downstream distance and trajectory estimates. This paper introduces *Localization Parity*, a fairness criterion that requires equal expected Intersection over Union across demographic groups conditional on detection. We present the first confounder-controlled audit spanning four driving datasets (BDD100K, CityPersons, EuroCity-Day, and EuroCity-Night), three detector families (YOLOX, RetinaNet, and Cascade R-CNN), and two demographic axes (gender and age), together with a demographic-weighted bounding box regression intervention. Female and child pedestrians receive significantly looser boxes in all twelve dataset and detector combinations, with mean gaps of 1.40 and 3.27 percentage points. These gaps survive scale stratification in 35 of 36 and 24 of 30 cells, respectively. The proposed intervention closes the training-set gap by 46.8%, yet fails to generalize to validation data. Localization Parity therefore exposes a previously unmeasured and safety-relevant axis of algorithmic unfairness, and the resulting fairness generalization gap poses a concrete open problem for equitable perception systems.

Keywords: Localization Parity · Algorithmic Fairness · Pedestrian Detection · Bounding Box Regression · Demographic Bias · Object Detection

1 Introduction

Pedestrian detection is central to modern computer vision pipelines for self-driving cars, surveillance, and traffic infrastructure. The cost of error is not

shared evenly across demographic groups. Prior work has shown that *detection recall* is unequal: Wilson et al. [14] report lower average precision for darker-skinned pedestrians, while Brandao [2] and Li et al. [9] find substantially higher miss rates for children than for adults across many detectors.

What these audits leave unmeasured is *localization quality*. Once a detector draws a box, the tightness of that box shapes every estimate that follows: position, distance to the ego vehicle, predicted trajectory, and braking urgency. Two detections can both pass the conventional $\text{IoU} \geq 0.5$ hit threshold and still differ sharply in geometry. A box at IoU 0.51 and a box at IoU 0.91 look identical under recall-based fairness metrics, yet they feed very different signals into downstream modules. If one demographic group systematically receives looser boxes, that group systematically receives worse downstream estimates, and no current fairness benchmark would notice.

We formalize this overlooked criterion as **Localization Parity (LP)**: equality of expected Intersection over Union (IoU) across demographic groups, conditional on successful detection and after controlling for confounders such as object scale. We then run the first multi-dataset, multi-detector LP audit and test a training-time mitigation.

Contributions.

- A formal definition of Localization Parity as a group fairness criterion on box regression quality, orthogonal to detection parity and linked to existing fairness notions for classification and regression.
- The first confounder-controlled LP audit across four autonomous driving datasets, three detector families spanning single-stage and two-stage designs, and two demographic axes: gender (female vs. male) and age (child vs. adult).
- Empirical evidence that female and child pedestrians receive consistently lower IoU boxes (mean gaps -1.40 pp and -3.27 pp), surviving ICON² scale stratification [13].
- Demographic-Weighted Bounding Box Regression (DWBBR), which closes the training gap by 46.8% yet fails to transfer cleanly to held-out data, documenting a *fairness generalization gap* as an open problem.

2 Related Work

Fairness research in machine learning has developed a mature vocabulary of group criteria, including demographic parity and equalized odds [7], individual fairness, and calibration. These notions were originally formulated for classification and ranking, where the outcome is typically binary or discrete. When the same ideas are imported into computer vision, they are usually operationalized through equal detection rates or equal average precision. Wilson et al. [14] established predictive inequity across Fitzpatrick skin-tone groups on BDD100K using AP, showing that standard detectors can under-serve darker-skinned pedestrians even after controlling for some scene factors. Schumann et al. [12] further

caution that binary third-party gender labels can themselves encode annotation bias, which limits how far any audit built on such labels can be trusted. Fair regression [1] offers a complementary continuous view by constraining disparities in expected prediction loss across groups. Localization Parity follows that continuous-loss perspective, but applies it specifically to the IoU of matched detections. What this broader fairness literature still lacks is a criterion that isolates box geometry after a detection has already been accepted.

Pedestrian detection fairness has likewise focused on hit-or-miss outcomes. BDD100K [15], CityPersons [16], and EuroCity Persons [3] remain the dominant driving benchmarks and supply the geographic and lighting diversity used in most recent audits. Brandao [2] evaluated twenty-four detectors on Caltech and reported substantially higher miss rates for children than for adults. Li et al. [9] broadened that finding across eight driving detectors and additionally observed higher female miss rates under low light. Khoshkdahan et al. [8] moved beyond overall accuracy to pose and occlusion as structural confounders of detection performance. Together these studies map the recall side of demographic bias carefully, yet they treat localization as a binary IoU threshold rather than as a continuous quality signal. As a result, two groups can appear equally “detected” while still receiving systematically different box tightness.

Detector design work has improved overall localization without targeting demographic parity. Cascade refinement [4] raises box quality through successive IoU thresholds, and focal reweighting [10] emphasizes hard classification examples, but neither objective equalizes IoU across protected groups. On the auditing side, ICON² [13] showed how to stratify by scale and occlusion so that AP gaps cannot be dismissed as size artifacts. On the training side, Group DRO [11] and related reweighting methods improve worst-group risk for classification. Both toolkits are valuable, yet they have been applied mainly to detection scores or class labels rather than to continuous box regression. The open question is therefore not whether demographic bias exists in pedestrian perception, but whether a localization-specific fairness gap exists, whether it survives confounder control, and whether a regression-level intervention can close it out of sample.

As Table 1 shows, existing audits stop at detection parity and leave continuous localization unexamined. None combine a multi-dataset, multi-detector IoU audit with scale stratification and a regression-level mitigation that is then checked out of sample. That missing combination is the research gap addressed in the remainder of this paper.

Table 1. Comparison of representative fairness audits in pedestrian detection. Prior work measures detection parity (miss rate or AP). This work measures continuous Localization Parity (mean IoU), adds confounder control, and tests a training-time mitigation.

Study	Measured outcome	out-	Dataset / detector scope	Scale control	Loc. mitigation
Wilson et al. [14]	AP		1 dataset	partial	no
Brandao [2]	miss rate		1 dataset; 24 detectors	no	no
Li et al. [9]	miss rate		several; 8 detectors	no	no
ICON ² [13]	AP		1 dataset	yes	no
Khoshkdahan et al. [8]	detection accuracy	accu-	1 dataset	yes	no
This work	mean (LP)	IoU	4 datasets; 3 detectors	yes	DWBBR

3 Proposed Solution

3.1 Localization Parity

Let $\mathcal{D} = \{(b_i, \hat{g}_i, g_i)\}$ be the set of successfully detected instances, where b_i is the predicted box, $\hat{g}_i$ the ground-truth box, and g_i the demographic group of instance i . Matching follows the COCO convention: a prediction counts as a detection only if $\text{IoU}(b_i, \hat{g}_i) \geq 0.5$. Localization Parity requires

$$\mathbb{E}[\text{IoU}(b_i, \hat{g}_i) \mid g_i = g_a] = \mathbb{E}[\text{IoU}(b_i, \hat{g}_i) \mid g_i = g_b] \quad (1)$$

for every pair of groups $a \neq b$, after controlling for instance scale. We report the signed gap $\Delta\text{IoU} = \mathbb{E}[\text{IoU} \mid g_{\text{priv}}] - \mathbb{E}[\text{IoU} \mid g_{\text{dis}}]$, where g_{priv} denotes the privileged group (male, adult) and g_{dis} the disadvantaged group (female, child). A negative gap means the disadvantaged group receives worse boxes. LP is deliberately conditional on detection: it equalizes box quality among people who were found, and is therefore orthogonal to detection parity. Relative to classification fairness criteria, LP replaces a binary outcome with a continuous quality score; relative to fair regression, it uses the loss $\ell = 1 - \text{IoU}$ on matched boxes; relative to AP-based audits, it isolates the localization component that AP mixes with confidence ranking.

Gaps of 1–3 pp IoU are practically consequential. Under a pinhole projection, relative depth error scales with relative box-height error. For a pedestrian at 30m whose true box height is 80px, a 2 pp mean IoU deficit (roughly 2–4 px of systematic height or vertical-center error) translates into about 0.7–1.5 m of depth bias. That is enough to alter time-to-collision estimates at urban speeds and to shift predicted crossing trajectories, so we treat LP gaps in this range as safety-relevant.

3.2 Demographic-Weighted Bounding Box Regression

DWBRR applies the Group DRO reweighting principle [11] to the CIoU regression branch. Standard YOLO training minimizes $L_{\text{class}} + \lambda L_{\text{reg}}$. DWBRR replaces the regression term with a group-weighted sum

$$L_{\text{DWBRR}} = L_{\text{class}} + \lambda \sum_i w_{g_i} L_{\text{CIoU}}(b_i, \hat{g}_i), \quad (2)$$

where $w_g \propto 1/\mathbb{E}_{\text{train}}[\text{IoU} \mid g]$ and weights are renormalized each epoch so that $\sum_g w_g = 1$. No architecture change is required: we override the Ultralytics training callback to recompute weights from per-group training CIoU and scale the regression loss before backpropagation.

3.3 Experimental Design

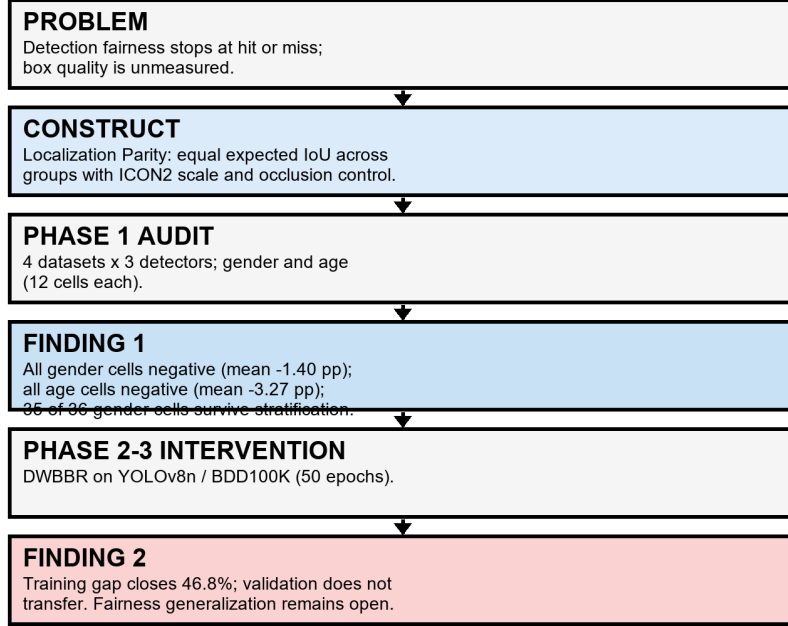


Fig. 1. Conceptual framework. Phase 1 audits Localization Parity across four datasets and three detectors under ICON² confounder control. Phase 2 and Phase 3 test whether DWBRR closes the gap and reveal a fairness generalization barrier.

Figure 1 is deliberate in sequencing measurement before mitigation. Without a confounder-controlled LP audit, a training intervention could appear to “fix fairness” while only compensating for size or lighting artifacts. By separating the twelve-cell audit from DWBBR, the framework forces any claimed fix to answer a harder question: does the gap shrink where the model has never trained?

We evaluate on four pedestrian benchmarks chosen for geographic, sensor, and lighting diversity: BDD100K [15] (large-scale US dashcam data with about 100,000 annotated pedestrian boxes and binary male/female labels), CityPersons [16] (about 19,000 boxes over 2,975 Cityscapes images with careful occlusion annotation), and EuroCity-Day together with EuroCity-Night [3] (subsets of EuroCity Persons, covering about 238,200 boxes from 31 European cities). Demographic labels for CityPersons and EuroCity use augmented annotations where available; detections with missing or ambiguous labels are dropped.

The audit covers three detector families with public pretrained weights and no fine-tuning: YOLOX [6] (single-stage, anchor-free), RetinaNet [10] (single-stage, anchor-based), and Cascade R-CNN [4] (two-stage, with cascaded IoU refinement). The mitigation experiments use YOLOv8n as a lightweight trainable backbone. The two protected attributes are gender (female vs. male) and age (child vs. adult). Race and skin tone are excluded because an earlier pilot lacked sufficient labeled sample size and used categories too coarse for reliable inference.

Predictions are matched to ground truth by greedy, confidence-sorted IoU assignment; matches with $\text{IoU} \geq 0.5$ enter the LP analysis. The primary confounder is object scale (box height), because female and child pedestrians tend to occupy smaller image regions. Following ICON², we bin ground-truth instances within each dataset into short (≤ 33 rd height percentile), medium (34th–66th), and tall (≥ 67 th) strata, then recompute per-group mean IoU inside each bin. Within each cell and stratum we compare group IoU distributions with the two-sided Mann–Whitney U test at $p < 0.05$.

Both DWBBR and an identically configured baseline (unweighted YOLOv8n) are trained on a 1,693-image BDD100K subset with reliable gender labels (64.8% male and 35.2% female), for 50 epochs, batch size 8, image size 416×416 , and $\text{lr}_0 = 0.01$ with cosine decay on a single NVIDIA RTX 3050. Validation uses a disjoint 423-image subset. We replicate with seeds $\{42, 123, 7\}$ and select the checkpoint with the best overall validation CIoU unless otherwise noted.

4 Experimental Results

4.1 Localization Parity Audit

Table 2 shows that the gender gap is not an artifact of one architecture: single-stage and two-stage detectors all under-localize female pedestrians, with a mean gap of -1.40 pp and significance in eleven of twelve cells. The CityPersons Cascade R-CNN exception (-0.49 pp, $p = 0.11$) is mild rather than reversing the sign, which suggests cascade refinement can shrink but not eliminate the disparity. The largest and most consistent deficits appear on BDD100K and EuroCity-

Table 2. Phase 1 LP audit, gender axis. Gap = female minus male mean IoU (pp). †: $p < 0.05$, Mann–Whitney U .

Dataset	Detector	Gap (pp)
BDD100K	YOLOX	$-2.16^\dagger$
BDD100K	RetinaNet	$-2.02^\dagger$
BDD100K	Cascade R-CNN	$-2.32^\dagger$
CityPersons	YOLOX	$-0.98^\dagger$
CityPersons	RetinaNet	$-0.71^\dagger$
CityPersons	Cascade R-CNN	-0.49
EuroCity-Day	YOLOX	$-0.67^\dagger$
EuroCity-Day	RetinaNet	$-0.65^\dagger$
EuroCity-Day	Cascade R-CNN	$-0.98^\dagger$
EuroCity-Night	YOLOX	$-1.81^\dagger$
EuroCity-Night	RetinaNet	$-2.47^\dagger$
EuroCity-Night	Cascade R-CNN	$-1.59^\dagger$
Mean		-1.40

Night, implying that scale diversity and low light amplify localization unfairness even when recall-based metrics would still count a hit.

Table 3. Phase 1 LP audit, age axis. Gap = child minus adult mean IoU (pp). †: $p < 0.05$, Mann–Whitney U .

Dataset	Detector	Gap (pp)
BDD100K	YOLOX	$-1.92^\dagger$
BDD100K	RetinaNet	-1.64
BDD100K	Cascade R-CNN	$-2.83^\dagger$
CityPersons	YOLOX	$-2.91^\dagger$
CityPersons	RetinaNet	$-3.11^\dagger$
CityPersons	Cascade R-CNN	$-3.48^\dagger$
EuroCity-Day	YOLOX	$-4.55^\dagger$
EuroCity-Day	RetinaNet	$-2.49^\dagger$
EuroCity-Day	Cascade R-CNN	$-3.17^\dagger$
EuroCity-Night	YOLOX	$-4.35^\dagger$
EuroCity-Night	RetinaNet	$-4.34^\dagger$
EuroCity-Night	Cascade R-CNN	$-4.44^\dagger$
Mean		-3.27

Table 3 reveals a sharper and more uniform failure mode for children: the mean age gap (-3.27 pp) is more than twice the gender gap, and every EuroCity-Night cell exceeds -4 pp. That pattern matters for safety because child pedestrians are already the group with elevated miss rates in prior recall audits; here

they also receive the loosest boxes when they *are* detected. In other words, detection parity and localization parity fail in the same direction, compounding rather than canceling risk.

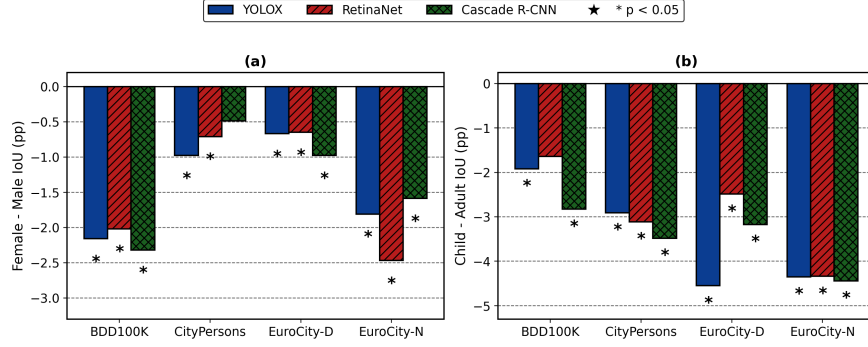


Fig. 2. Phase 1 LP audit: gender gaps (a) and age gaps (b) across four datasets and three detectors. Asterisks mark $p < 0.05$.

Figure 2 makes the structural takeaway visible at a glance: almost every bar lies below zero, so Localization Parity is violated as a systematic property of current detectors rather than as a rare edge case. Panel (b) also shows that age gaps dominate gender gaps on every dataset, while night scenes pull both axes downward. Under ICON² scale stratification the gender gap survives in 35 of 36 cells and the age gap in 24 of 30, confirming that the pattern in the figure is not explained by women and children simply occupying smaller image regions.

4.2 DWBBR Intervention

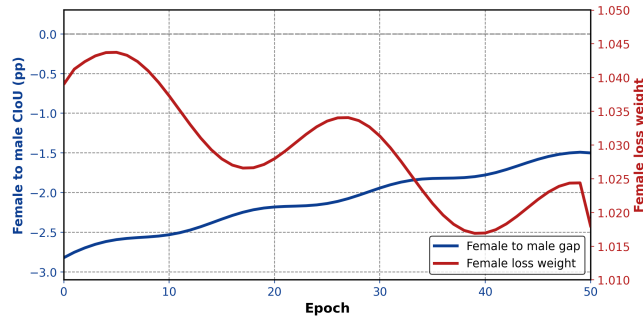


Fig. 3. DWBBR training dynamics: LP gap convergence over 50 epochs on BDD100K.

Figure 3 shows that the proposed reweighting is not inert: over 50 epochs the female-to-male training CIoU gap narrows from -2.82 pp to -1.50 pp (46.8% relative reduction) as the female loss weight drifts toward parity. This is evidence that Localization Parity is actionable inside the optimization loop—the regression branch *can* be steered—but it does not yet answer whether that steering survives the distribution shift to held-out frames.

Table 4. Validation CIoU by group on the BDD100K validation subset (423 images). Gap = weaker group minus reference group (pp). Multi-seed means use seeds {42, 123, 7}.

Group	Model	n	CIoU	95% CI	Gap (pp)
Male	Baseline	555	0.7942	—	—
Female	Baseline	376	0.7790	[0.768, 0.790]	-1.52
Male	DWBBR	555	0.7990	—	—
Female	DWBBR	374	0.7715	[0.760, 0.783]	-2.74
Mean female to male, baseline					-1.78 ± 0.48
Mean female to male, DWBBR					-1.36 ± 0.63

Table 4 exposes the fairness generalization gap directly. Under the primary seed, DWBBR slightly improves male CIoU while *worsening* female CIoU, widening the gap from -1.52 pp to -2.74 pp with no significant female gain ($p = 0.71$). Across three seeds the mean validation gap shrinks only 23.6% and becomes more variable, with best checkpoints scattered across epochs {15, 17, 21}. Closing Localization Parity on the training set is therefore not evidence that the same model will deliver equitable boxes at deployment.

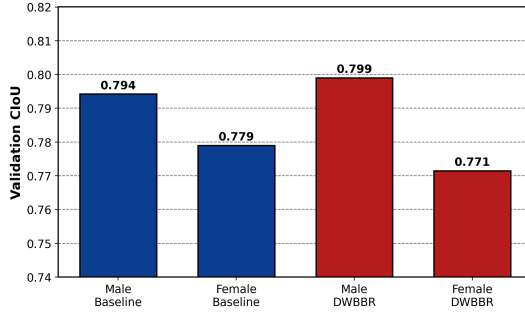


Fig. 4. Validation-set CIoU for DWBBR and the baseline, by demographic group.

Figure 4 visualizes the same tension: demographic reweighting does not lift the disadvantaged group above the baseline on held-out data. The plot under-

scores why mAP-only model selection is insufficient for fairness work—a checkpoint that looks competitive in aggregate CIoU can still leave female pedestrians with systematically looser boxes.



Fig. 5. Qualitative Localization Parity violations on BDD100K. Female pedestrians more often receive looser predicted boxes than male pedestrians in comparable scenes.

Figure 5 shows what the quantitative gaps look like in scene context: female pedestrians receive boxes that leave more background inside the prediction or crop the body more aggressively than nearby male pedestrians under similar scale and occlusion. These examples matter because downstream modules consume geometry, not group labels; a looser box is already a distorted depth and trajectory signal before any fairness metric is computed.

Demographic labels in BDD100K are third-party visual inferences with binary gender, no self-identification, and undocumented annotation guidelines, raising consent and inclusivity concerns [12]. LP is a proxy for box-quality fairness rather than a guarantee of equal downstream risk, and fairer localization can reduce collision risk while also improving re-identification. We publish because we judge the near-term safety benefit to outweigh that dual-use risk. The study examines only gender and age; race, skin tone, disability, body type, and intersectional analyses remain open.

5 Conclusion

Localization Parity addresses a previously unmeasured fairness gap in pedestrian detection: after a person is found, existing audits ignore how tightly the box fits, even though box geometry drives distance, trajectory, and braking estimates. This paper contributes a formal LP criterion, the first confounder-controlled multi-dataset and multi-detector audit of continuous IoU disparities on gender and age, and a demographic-weighted regression intervention. Across four

datasets and three detector families, female and child pedestrians receive systematically lower IoU boxes (mean gaps -1.40 pp and -3.27 pp), and the gaps survive scale stratification, confirming that the disparity is not explained by object size alone.

Future work should evaluate DWBBR across datasets and architectures, adopt fairness-aware checkpoint selection, compare stronger DRO and oversampling baselines, extend audits to skin tone and intersectional groups with higher-quality labels, and quantify how 1–3 pp IoU gaps propagate into planning decisions.

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